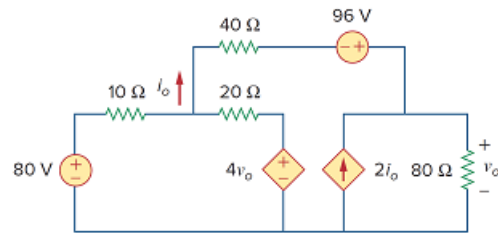


Problem

Using nodal analysis, find v_o and i_o in the circuit of Fig. 3.79.

Figure 3.79:



Step-by-step solution

Step 1 of 7

Draw the circuit diagram shown in Figure 1.

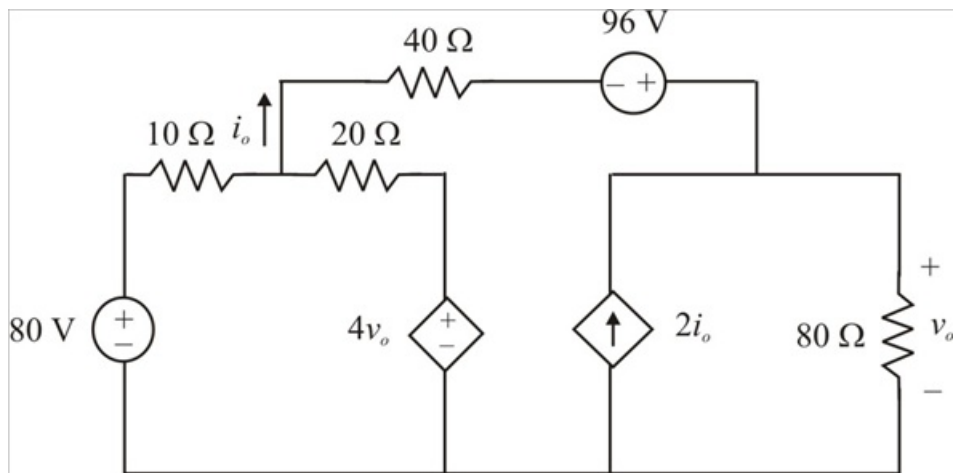


Figure 1

Step 2 of 7

The circuit is labeled as shown in Figure 2.

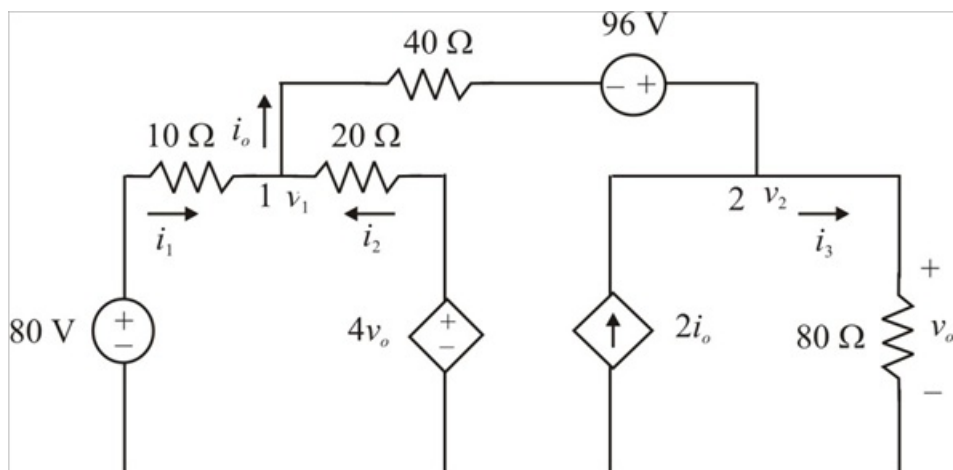


Figure 2

Step 3 of 7

Applying Kirchhoff's current law at node 1, we get,

$$i_1 + i_2 = i_o$$

$$\frac{80 - v_1}{10} + \frac{4v_o - v_1}{20} = \frac{v_1 + 96 - v_2}{40}$$
$$320 - 4v_1 + 8v_o - 2v_1 = v_1 + 96 - v_2$$
$$7v_1 - v_2 - 8v_o = 224$$

But $v_o = v_2$.

Substitute v_2 for v_o .

$$7v_1 - v_2 - 8v_o = 224$$

$$7v_1 - v_2 - 8v_2 = 224$$

$$7v_1 - 9v_2 = 224$$

Therefore $7v_1 - 9v_2 = 224$ (1)

Step 4 of 7

Applying Kirchhoff's current law at node 2, we get

$$i_o + 2i_o = i_3$$

$$3i_o = \frac{v_o}{80}$$

$$3\left(\frac{v_1 + 96 - v_2}{40}\right) = \frac{v_2}{80}$$

$$6v_1 + 576 - 6v_2 = v_2$$

$$6v_1 - 7v_2 = -576$$

Therefore $6v_1 - 7v_2 = -576$ (2)

Step 5 of 7

Multiplying equation (1) by 6 and equation (2) by 7 and subtract equation (2) from it, we get

$$6(7v_1 - 9v_2 = 224)$$

$$7(6v_1 - 7v_2 = -576)$$

$$42v_1 - 54v_2 = 1344$$

$$42v_1 - 49v_2 = -4032$$

$$\hline -5v_2 = 5376$$

$$v_2 = -1075.2 \text{ V}$$

But $v_o = v_2$, hence,

$$v_o = v_2$$

$$= -1075.2 \text{ V}$$

Therefore, the voltage v_o is -1075.2 V .

Step 6 of 7

Substituting $v_2 = -1075.2 \text{ V}$ in equation (2), we get,

$$6v_1 - 7v_2 = -576$$

$$6v_1 - 7(-1075.2) = -576$$

$$6v_1 + 7526.4 = -576$$

$$6v_1 = -8102.4$$

$$v_1 = -1350.4 \text{ V}$$

Step 7 of 7

The current i_o can be calculated as:

$$\begin{aligned} i_o &= \frac{v_1 + 96 - v_2}{40} \\ &= \frac{-1350.4 + 96 - (-1075.2)}{40} \\ &= \frac{-1350.4 + 96 + 1075.2}{40} \\ &= -4.48 \text{ A} \end{aligned}$$

Therefore, the current i_o is -4.48 A.